

# SHREEJIT VERMA

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## EDUCATION

**Georgia Institute of Technology (Online)**, Atlanta, USA

Aug 2024 – Dec 2026 (exp.)

Master of Science in Computer Science with Specialization in Computing Systems

Coursework: Computer Networks, Advanced Operating Systems, Distributed Computing, Database Management Systems

**Stevens Institute of Technology**, Hoboken, NJ

Aug 2024 – May 2026

Master of Science in Financial Engineering | **GPA:** 3.974/4.0

Coursework: Market Microstructure, Portfolio Theory and Applications, Algorithmic Trading Strategies, Multivariate Statistics

**WorldQuant University**, New Orleans, USA

Dec 2021 – May 2024

Master of Science in Financial Engineering | **GPA:** 86%

Coursework: Deep Learning for Finance, Financial Econometrics, Fixed Income, Equity, Portfolio Management, Risk Management

**Carnegie Mellon University, Tepper School of Business**, New York, USA

Aug 2021 – Oct 2021

Master of Science in Computational Finance – (Program withdrawn due to father's illness)

Coursework: Investments, Statistical Machine Learning, Simulation Methods, Financial Computing, Algorithmic Optimization

**Vellore Institute of Technology**, Vellore, India

Bachelor of Technology in Computer Science and Engineering | **GPA:** 8.78/10.0

July 2014 – Sept 2018

Coursework: Data Structures and Algorithms, Programming Language Translators, Natural Language Processing (NLP)

## PROFESSIONAL EXPERIENCE

**BNP Paribas CIB**, New York, USA

*C++ Quantitative Developer (Co-op), Automated Market Making*

Feb 2026 – May 2026

- Built low-latency components of the automated market-making stack for the Prime Credit Market (average **\$500M** of daily market-making volume), spanning real-time market-data ingestion, tick analytics, and pricing/execution paths
- Developed Python services for trading analytics, risk/trade monitoring, and creating reliable research and operational tooling
- Integrated secure on-premise LLM tooling with Git/Jira/Confluence to automate code, testing, and documentation workflows

**LogiNext Solutions Inc.**, Mumbai, India

*Senior Software Engineer, Analytics*

Mar 2023 – July 2024

- Architected and developed Map Construction Algorithms, Map Routing Algorithms, and solved Rich Vehicle Routing Problems (3 Nested NP-Hard Problems) using CP-SAT Constraint Programming and Convex optimization using Python, PostGIS, and S3
- Led a 12-engineer team delivering a high-throughput geospatial mapping application platform
- Built an LLM-powered debugging and query-resolution tool across the company, cutting mean bug-resolution time by **80%**

**Versor Investments (QR Systems LLP)**, Mumbai, India

*Quantitative Developer, Merger Arbitrage and Stock Selection Portfolio*

Feb 2022 – Oct 2022

- Developed and backtested systematic merger-arbitrage strategies of an **\$8.5 Billion** AUM fund, improving alpha capture by **15%**
- Built and deployed ML pipelines for Order and Execution Management Systems, increasing trade execution efficiency by **29%**
- Designed an ESG-driven merger-arbitrage signal capitalizing on arbitrage opportunities on pre- and post-merger statistics

**Bank of America (BA Continuum India Pvt Ltd.)**, Chennai, India

*Senior Software Engineer, Fixed Income Commodities and Currencies (FICC)*

Jan 2020 – July 2021

- Engineered Python-based trading services enhancing the storage, processing, matching, and execution of trades on QUARTZ
- Integrated C++ pipelines to store trades in the object-oriented database SANDRA, reducing trade processing latency by **50%**
- Led the migration of **1 million** plus lines of code to Python 3.8, enhancing system scalability and execution efficiency by **40%**

*Senior Tech Associate in Data Analysis and Insight Technology*

June 2018 – Dec 2019

- Architected and developed an ML/AI platform to deploy predictive models, increasing decision-making accuracy by **67%**
- Designed machine learning models for data validation rules prediction, reducing by close to **36 Full-Time Equivalents (FTEs)**

## SKILLS

**Mathematics/Statistics:** Probability, Stochastic Calculus, Differential Equations, PDE, Linear Algebra, Numerical Methods, Markov

**Quantitative Finance:** Statistical Analysis, Derivative Pricing, Time Series Analysis, Factor Modeling, Predictive Modeling, Greeks

**Machine Learning:** Linear Regression, Clustering, Random Forest, XGBoost, RNN, LSTM, Deep Learning, Neural Networks, NLP

**Programming:** C++ (17/20/23, primary), Python, C, Java, R, MATLAB, JavaScript, Node.js, ReactJS, NumPy, Pandas, Polars,

SciPy, Keras, PyTorch, TensorFlow, Scikit-learn, QuantLib, Statsmodels, QtPy, CVXPY, OpenMP, MPI, CUDA, Bash Scripting

**Data Engineering:** Airflow, Dask, Spark, PySpark, FastAPI, REST APIs, Kafka, Flink, NoSQL, GraphQL, SQL, BQL, KDB+, Q,

VBA, ETL, Big Data, Hadoop, HDFS, PostgreSQL, MongoDB, ZeroMQ, Cassandra, Django, Redis, JIRA, Confluence, Agile, Scrum

**Computer Science:** TCP/IP, UDP, Multicast, Cache and Multithreading optimization, FPGA (Verilog, VHDL), Blockchain

**Cloud Computing and DevOps:** Linux, Git, Jenkins, CI/CD, Ansible, Docker, Kubernetes, Helm, AWS, GCP, WSL

## RESEARCH & SELECTED PROJECTS

**AI-Integrated FPGA for Market Making in Volatile Environments (MS Thesis, Stevens)** | [GitHub](#)

Oct 2025 – May 2026

- Engineered a sub-10 $\mu$ s low-latency trading system, with a custom-built limit order book, FPGA market data handlers, kernel bypass (DPDK), hardware timestamping, and lock-free data structures for deterministic, microsecond-level execution

**Sovereign AI Command Center** | [Website](#)

May 2026 – Aug 2026

- Architected a local-first, autonomous agentic AI platform orchestrating LLM providers behind a unified API, enabling agents to autonomously execute multi-step tool-calling workflows via the Model Context Protocol (MCP) with zero data leaving the host.

- Developed a Retrieval-Augmented Generation (RAG) and persistent semantic-memory system on ChromaDB using high-dimensional vector embeddings for low-latency retrieval, plus a hardware-aware layer that deploys quantized open-weight models
- Plug-and-Play Agentic AI Engineering Harness** | [GitHub](#) May 2026 – July 2026
- Architected a cross-platform agentic AI developer platform using NixOS, declarative configuration, version-pinned dependencies, symlink-managed files, and automated health checks to bootstrap a complete local AI engineering environment on clean machines
  - Integrated multi-agent orchestration and agent-native CLI workflows to automate isolated Git worktrees, autonomous task execution, CI-gated shipping (code review/testing/linting, PR in one command), overnight runs, and upstream synchronization
- Adaptive Volatility Regime-Based Execution and Risk Framework** | [GitHub](#) Sept 2025 – Dec 2025
- Developed adaptive volatility regime switch framework dynamically selecting among passive, TWAP, and aggressive strategies
  - Achieved **20.0%** increase in Sharpe Ratio, **6.1%** transaction cost reduction, **20.1%** CVaR decrease with robust risk management
- Statistical Arbitrage Reversal and Momentum Strategies (Quant Researcher, WallStreetQuants)** | [GitHub](#) Jun 2025 – Aug 2025
- Designed and backtested a 120-day volume-momentum-based crypto portfolio strategy, yielding a **155.76%** annualized return and a **1.94** Sharpe Ratio (post transaction costs), significantly outperforming the Bitcoin buy-and-hold benchmark
- Dynamic Portfolio Optimization (Master's Thesis, MScFE, WorldQuant University)** | [GitHub](#) Mar 2024 – June 2024
- Built a real-time portfolio optimization system using convex and non-convex optimization methods, enhancing risk-adjusted returns via adaptive asset rebalancing and multi-factor modeling, managing interest rate, FX, credit, and market risks
- Environmental Social Governance (ESG) Merger Arbitrage Strategy** Apr 2022 – June 2022
- Developed the ESG Strategy, which was subsequently converted into a portfolio and embedded across all existing portfolios. It caters to the arbitrage opportunity being made by the effect of ESG scores on target and acquirer pre- and post-merger statistics
- Blockchain In Retail** Jan 2018 – Mar 2018
- Developed a decentralized e-commerce platform to secure and streamline retail transactions with smart contracts, currency conversion, custom hashing, and matching algorithms. Using Homebrew, Truffle, MetaMask, Ganache, GETH, Solc, Puppeth
- QS Rank Predictor** June 2017 – July 2017
- Constructed an ensemble machine learning model consisting of multiple Deep neural networks to predict QS World Ranking
  - The model also gave suggestions on areas to improve, helping in achieving a world ranking of 301 - 400 for the VIT CS in 2020

## ACHIEVEMENTS

**1st Place** for personal portfolio in Vanguard ETF Trading Challenge | **President** of Stevens Graduate Financial Association | State Rank Holder in **International Science Olympiad** and **International Mathematics Olympiad** | **Beta Gamma Sigma** Member

### Global Recognition Gold Award (Bank of America)

- Led enterprise-wide AI/ML campaign identifying 64 high-impact use cases on fraud detection, forecasting, and optimization
- Organized 4 large-scale Data Science events across Bank of America, delivering AI/ML lectures to 2500+ employees

### Global Recognition Silver Award (Bank of America, 2x)

- For contribution in Total Return Swap Bonds, Futures, Options, and Cash (LOBs) under the Post Trade Processing Team
- Architected and developed an AI/ML framework integrating end-to-end tools for building in-house Data Science Projects

## CERTIFICATIONS

\*Links to certificates are attached as hyperlinks to the respective course names

- **Finance:** CFA Level 1; Bloomberg Market Certification; Financial Engineering and Risk Management Part I & II) (Coursera); Investment Foundations Program (CFA, USA); **The Complete Financial Analyst Training & Investing Course** (Udemy); **Machine Learning for Trading Specialization** (Google Cloud Platform, New York Institute of Finance): (Introduction to Trading, Machine Learning & GCP, Using Machine Learning in Trading and Finance, Reinforcement Learning for Trading Strategies); **Investment Management Specialization** (University of Geneva, UBS): (Understanding Financial Markets, Meeting Investors' Goals, Portfolio and Risk Management, Securing Investment Returns in the Long Run, Planning your Client's Wealth over a 5-year Horizon); **Trading Strategies in Emerging Markets Specialization** (Indian School of Business, ISB): (Trading Basics, Trading Algorithms, Advanced Trading Algorithms, Creating a Portfolio, Design your own trading strategy – Culminating Project); **Finance & Quantitative Modeling for Analysts Specialization** (University of Pennsylvania, Wharton): (Fundamentals of Quantitative Modeling, Introduction to Spreadsheets and Models, Financial Acumen for Non-Financial Managers, Introduction to Corporate Finance); **Corporate Finance and Valuation** (NYU STERN, Aswath Damodaran)
- **Computer Science:** **Deep Learning Specialization:** (Statistical Inference, Regression Models, Practical Machine Learning, Developing Data Products, Data Science Capstone); **Applied Data Science with Python Specialization** (University of Michigan): (Introduction to Data Science in Python, Applied Plotting, Charting & Data Representation in Python, Applied Machine Learning in Python, Applied Text Mining in Python, Applied Social Network Analysis in Python); **Data Science Foundations using R Specialization** (Johns Hopkins University): (The Data Scientist's Toolbox, R Programming, Getting and Cleaning Data, Exploratory Data Analysis, Reproducible Research); **Data Science Statistics and Machine Learning Specialization:** (Statistical Inference, Regression Models, Practical Machine Learning, Developing Data Products, Data Science Capstone); **Big Data Specialization** (University of California San Diego): (Introduction to Big Data, Big Data Modeling and Management Systems, Big Data Integration and Processing, Machine Learning with Big Data, Graph Analytics for Big Data, Big Data - Capstone Project); **Data Structures and Algorithms Specialization:** (Algorithmic Toolbox, Data Structures, Algorithms on Graphs, Algorithms on Strings, Advanced Algorithms and Complexity, Genome Assembly Programming Challenge); **Algorithms, Part I & Part II** (Princeton University)

**Interests:** Chess, Poker, F1, Martial Arts, Cricket, Boxing, Badminton, Reading, Cooking, Dancing, Psychology, History, Philosophy